GATE ME 90 Days Challenge

Day 33/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
General Aptitude	Analytical Reasoning	73 days to GATE

How to use this set: 15 questions (9 MCQ, 6 NAT) on syllogisms, if-then logic, number and letter series, coding, blood relations, directions, clocks and calendars, data sufficiency, seating arrangement, ranking and figure counting. Try them in **40 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Statements: No engineer is lazy. All managers are engineers.
Conclusions: I. No manager is lazy. II. Some engineers are managers.
Which conclusion(s) follow(s)?

(A) Only I
(B) Only II
(C) Both I and II
(D) Neither I nor II

Q2 MCQ

"The pump delivers water only if it is primed." The pump delivered water this morning. Which conclusion is certain?

(A) The pump was primed this morning
(B) The pump was not primed this morning
(C) A primed pump always delivers water
(D) Nothing can be concluded

Q3 NAT

The next term in the series 2, 6, 12, 20, 30, ___ is _____

Q4 MCQ

Find the next letter in the series: B, E, I, N, ___

(A) S
(B) T
(C) U
(D) R

Q5 NAT

In a certain code, GATE = 33 and MECH = 29. In the same code, PUMP = _____

Q6 MCQ

A is B's brother. C is A's mother. D is C's father. How is A related to D?

(A) Son
(B) Grandson
(C) Nephew
(D) Great-grandson

Q7 NAT

A person walks 5 km east, turns left and walks 3 km, turns left again and walks 9 km, then turns left again and walks 6 km. His distance from the starting point (in km) is _____

Q8 NAT

The hands of a clock coincide for the first time after 3 o'clock at how many minutes past 3 (round off to two decimal places)? _____

Q9 MCQ

1 January 2027 is a Friday. What day of the week is 1 March 2028?

(A) Tuesday
(B) Wednesday
(C) Thursday
(D) Friday

Q10 MCQ

Question: Is $x > y$, where x and y are real numbers?
Statement I: $x^2 > y^2$
Statement II: $x + y > 0$

(A) Statement I alone is sufficient
(B) Statement II alone is sufficient
(C) Both statements together are sufficient, but neither alone is
(D) Both statements together are not sufficient

Q11 MCQ

Six friends A, B, C, D, E and F sit around a circular table facing the centre. A sits opposite D. B sits immediately to the right of A. C is not an immediate neighbour of D. E does not sit opposite B. Who sits opposite C?

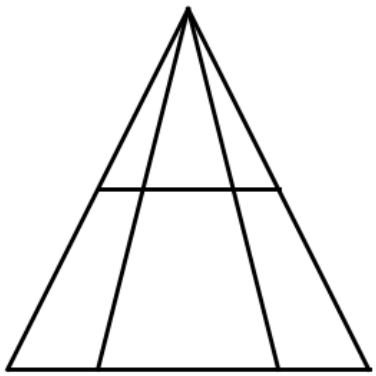
(A) B
(B) E
(C) F
(D) A

Q12 NAT

Ravi is 14th from the top and 27th from the bottom in his class. The number of students in the class is _____

Q13 NAT

The number of triangles in the figure shown is _____



Q14 MCQ

Statements: Some bolts are nuts. All nuts are screws.
Conclusions: I. Some bolts are screws. II. All screws are nuts.
Which conclusion(s) follow(s)?

(A) Only I
(B) Only II
(C) Both I and II
(D) Neither I nor II

Q15 MCQ

One of three suspects A, B and C broke a gauge. A says, "B did it." B says, "I did not do it." C says, "A did not do it." Exactly one of the three statements is true. Who broke the gauge?

(A) A
(B) B
(C) C
(D) Cannot be determined

Answer Key

Q	Type	Answer
Q1	MCQ	(C) Both I and II
Q2	MCQ	(A) The pump was primed this morning
Q3	NAT	42
Q4	MCQ	(B) T
Q5	NAT	66
Q6	MCQ	(B) Grandson
Q7	NAT	5
Q8	NAT	16.36 (accept 16.30 to 16.40)
Q9	MCQ	(B) Wednesday
Q10	MCQ	(C) Both statements together are sufficient, but neither alone is
Q11	MCQ	(B) E
Q12	NAT	40
Q13	NAT	12
Q14	MCQ	(A) Only I
Q15	MCQ	(A) A

Detailed Solutions

Q1 **Answer: (C) Both I and II**

Concept: Draw the Venn diagram: the managers circle lies inside the engineers circle, which is completely separate from the lazy circle.

I: managers sit inside engineers, so none of them can be lazy → follows. II: "All managers are engineers" converts to "Some engineers are managers" → follows.

Answer: both I and II.

Exam tip: "All A are B" converts only to "Some B are A", never to "All B are A".

Q2 **Answer: (A) The pump was primed this morning**

Concept: "P only if Q" means "if P then Q": delivery $\Longrightarrow$ primed.

The pump delivered water (P is true), so it **was primed** (Q is true).

Option (C) is the converse (Q $\Longrightarrow$ P), which is not implied.

Exam tip: "only if" introduces the necessary condition; "if" introduces the sufficient condition.

Q3 **Answer: 42**

Concept: Look at the differences: 4, 6, 8, 10, so the next difference is 12.

Next term = 30 + 12 = **42** (the terms are n(n + 1): 1×2, 2×3, ..., 6×7).

Exam tip: If first differences form an AP, the series is quadratic in n.

Q4 **Answer: (B) T**

Concept: Convert letters to positions: B = 2, E = 5, I = 9, N = 14.

Gaps are +3, +4, +5, so the next gap is +6: 14 + 6 = 20 = **T**.

Exam tip: Remember the EJOTY markers: E = 5, J = 10, O = 15, T = 20, Y = 25.

Q5 **Answer: 66**

Concept: The code is the sum of the alphabet positions of the letters.

GATE = 7 + 1 + 20 + 5 = 33 ✓; MECH = 13 + 5 + 3 + 8 = 29 ✓

PUMP = 16 + 21 + 13 + 16 = **66**

Exam tip: When a number code is given for a word, test the position sum, the product and the reverse positions (27 − n) first.

Q6 **Answer: (B) Grandson**

Concept: Build the family tree generation by generation.

D is C's father; C is A's mother, so D is A's maternal grandfather. A is male (a brother), so A is D's **grandson**.

Exam tip: Fix the gender of each person as soon as a word like brother, mother or father reveals it.

Q7 **Answer: 5**

Concept: Track the coordinates (east = +x, north = +y).

East 5 → (5, 0); left = north 3 → (5, 3); left = west 9 → (−4, 3); left = south 6 → (−4, −3).

Distance = $\sqrt{(4^2 + 3^2)}$ = **5 km** (towards the south-west).

Exam tip: Three left turns in a row bring you back to facing the original direction's right.

Q8 **Answer: 16.36 (accept 16.30 to 16.40)**

Concept: The minute hand gains 5.5° per minute on the hour hand.

At 3:00 the hour hand is 90° ahead. Time to close the gap = 90/5.5 = 180/11 min

= 16 4/11 min = **16.36 min** past 3.

Exam tip: The hands coincide 11 times in 12 hours, every 65 5/11 minutes.

Q9 **Answer: (B) Wednesday**

Concept: Count odd days (days beyond complete weeks).

1 Jan 2027 → 1 Jan 2028: 365 days = 1 odd day → Saturday.

1 Jan 2028 → 1 Mar 2028: January 31 + February 29 (2028 is a leap year) = 60 days = 4 odd days → Saturday + 4 = **Wednesday**.

Exam tip: A century year is a leap year only if divisible by 400; other years if divisible by 4.

Q10 **Answer: (C) Both statements together are sufficient, but neither alone is**

Concept: Test each statement with counter-examples, then combine.

I alone: $x = -3$, $y = 1$ gives $x^2 > y^2$ but $x < y$; $x = 3$, $y = 1$ gives $x > y$. Not sufficient.

II alone: $x = 1$, $y = 2$ (no) and $x = 2$, $y = 1$ (yes). Not sufficient.

Together: $x^2 - y^2 = (x - y)(x + y) > 0$ with $x + y > 0$ forces $x - y > 0$. **Both together are sufficient.**

Exam tip: Factorising is often the fastest way to combine two inequality statements.

Q11 **Answer: (B) E**

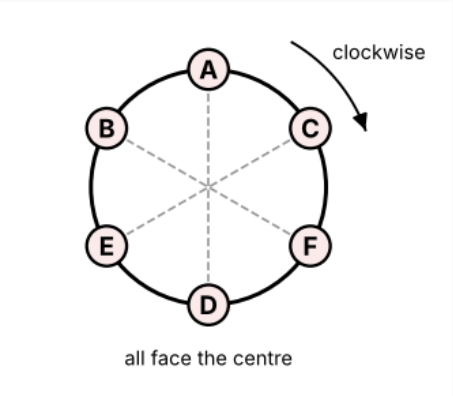
Concept: Fix A, then place the others clockwise (facing the centre, a person's right neighbour is the next seat anticlockwise).

Seats clockwise from A: A, 2, 3, D, 5, B (B is to A's right, D is opposite A).

C cannot be next to D (seats 3 and 5), so C = seat 2. E is not opposite B (seat 3), so E = seat 5 and F = seat 3.

Clockwise order: A, C, F, D, E, B. Opposite pairs: A–D, C–E, F–B. So **E** is opposite C.

Exam tip: In a circle facing the centre, left = clockwise and right = anticlockwise; facing outward, they swap.



Q12 **Answer: 40**

Concept: Total = rank from top + rank from bottom − 1 (Ravi is counted twice).

Total = 14 + 27 − 1 = **40**

Exam tip: Between-two-people questions: number between = |difference of positions| − 1.

Q13 **Answer: 12**

Concept: Every triangle here has its apex at the top vertex; its base is any segment between two of the four points on a horizontal line.

On the base line there are 4 points (the two corners and the two feet of the inner lines): C(4, 2) = 6 triangles.

On the middle horizontal line there are also 4 points: C(4, 2) = 6 triangles.

Total = 6 + 6 = **12**

Exam tip: For n lines from one apex (counting the two sides) cut by k horizontal lines (counting the base), the count is k × C(n, 2) = 2 × C(4, 2) here.

Q14 **Answer: (A) Only I**

Concept: The bolts that are nuts lie inside the screws circle.

So some bolts are screws: I follows. "All nuts are screws" does not mean all screws are nuts: II does not follow.

Answer: only I.

Exam tip: Some + All (in that order) gives a "Some" conclusion; All + Some gives no definite conclusion.

Q15 **Answer: (A) A**

Concept: Try each suspect as the culprit and count the true statements.

If A did it: A's statement false, B's true, C's false → exactly one true ✓.

If B did it: A's true, B's false, C's true → two true ✗. If C did it: A's false, B's true, C's true → two true ✗.

So **A** broke the gauge.

Exam tip: With a small number of cases, test every case systematically instead of reasoning in circles.

What next?

Day 33 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

Telegram: daily PDFs

Next, Day 34/90: Fluid Mechanics, Turbulent Flow & Pipe Losses. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.